

## MARCH AFB, CA, USA

WMO: 722860

Lat: 33.900N

Lon: 117.250W

Elev: 468

StdP: 95.83

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23119

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | -0.3       | 1.3 | -17.1                         | 0.9 | 16.1 | -14.1 | 1.2 | 16.5 | 11.1                  | 12.6 | 9.3 | 12.6 | 0.6                   | 320  | 0.435 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 17.6                   | 38.5            | 19.8 | 37.1 | 19.3 | 35.5 | 19.0 | 22.2                | 34.0 | 21.4 | 32.9 | 20.6 | 31.7 | 4.6                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 14.7 | 24.1 | 17.9 | 13.6 | 23.4 | 17.2 | 13.0 | 22.8 | 67.0          | 33.8 | 64.0 | 32.7 | 61.4 | 31.7 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |      |     |
| (4) 8.0           | 7.0  | 5.9 | DB                         | -3.6 | 41.8               | 2.5 | 1.5                                                | -5.5 | 42.9       | -6.9 | 43.8       | -8.3 | 44.6       | -10.1 | 45.7 | (4) |
| (5)               |      |     | WB                         | -5.1 | 23.7               | 2.2 | 1.1                                                | -6.7 | 24.4       | -8.0 | 25.1       | -9.2 | 25.7       | -10.8 | 26.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.7   | 11.6  | 11.5  | 13.7  | 15.4  | 18.4  | 22.0  | 25.4  | 26.1  | 23.9  | 19.0  | 14.4  | 11.0 |
|                                                                                  | DBStd       | 6.21   | 3.16  | 3.23  | 3.12  | 3.43  | 3.39  | 3.07  | 2.60  | 2.84  | 3.44  | 3.34  | 3.34  | 3.03 |
|                                                                                  | HDD10.0     | 77     | 20    | 18    | 5     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 26   |
|                                                                                  | HDD18.3     | 1080   | 210   | 191   | 146   | 99    | 41    | 6     | 0     | 0     | 2     | 32    | 124   | 229  |
|                                                                                  | CDD10.0     | 2896   | 68    | 61    | 121   | 164   | 260   | 359   | 477   | 498   | 416   | 279   | 136   | 56   |
|                                                                                  | CDD18.3     | 857    | 1     | 1     | 3     | 12    | 42    | 115   | 218   | 240   | 168   | 52    | 6     | 0    |
|                                                                                  | CDH23.3     | 11988  | 52    | 66    | 181   | 387   | 728   | 1583  | 2719  | 3021  | 2089  | 890   | 244   | 28   |
|                                                                                  | CDH26.7     | 6377   | 6     | 10    | 44    | 140   | 326   | 821   | 1558  | 1807  | 1178  | 411   | 76    | 1    |
| Wind                                                                             | WSAvg       | 2.4    | 2.1   | 2.3   | 2.2   | 2.6   | 2.7   | 2.7   | 2.7   | 2.5   | 2.3   | 2.2   | 2.0   | 2.2  |
| Precipitation                                                                    | PrecAvg     | 294    | 60    | 75    | 40    | 18    | 6     | 2     | 4     | 2     | 5     | 11    | 25    | 44   |
|                                                                                  | PrecMax     | 565    | 288   | 284   | 168   | 62    | 51    | 18    | 38    | 10    | 38    | 140   | 54    | 215  |
|                                                                                  | PrecMin     | 103    | 2     | 2     | 0     | 1     | 0     | 0     | 1     | 0     | 0     | 0     | 6     | 4    |
|                                                                                  | PrecStd     | 134    | 74    | 64    | 37    | 14    | 10    | 4     | 7     | 2     | 8     | 26    | 13    | 45   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.4  | 28.2  | 30.7  | 33.7  | 36.3  | 39.1  | 40.0  | 40.8  | 39.8  | 36.4  | 31.8  | 26.1 |
|                                                                                  |             | MCWB   | 12.4  | 12.7  | 14.8  | 14.9  | 17.8  | 18.8  | 20.9  | 20.6  | 19.3  | 16.5  | 14.0  | 11.4 |
|                                                                                  | 2%          | DB     | 24.8  | 25.5  | 27.6  | 30.3  | 33.0  | 35.9  | 37.4  | 38.7  | 37.4  | 33.6  | 29.0  | 23.8 |
|                                                                                  |             | MCWB   | 11.2  | 11.9  | 14.1  | 14.5  | 16.8  | 18.6  | 20.0  | 19.8  | 19.1  | 15.4  | 13.2  | 10.6 |
|                                                                                  | 5%          | DB     | 22.4  | 22.5  | 25.2  | 27.8  | 30.5  | 33.7  | 35.9  | 37.1  | 35.5  | 31.4  | 26.3  | 21.3 |
|                                                                                  |             | MCWB   | 10.6  | 11.2  | 13.3  | 13.9  | 16.1  | 18.3  | 19.8  | 19.6  | 18.8  | 15.0  | 12.2  | 9.8  |
|                                                                                  | 10%         | DB     | 19.9  | 19.4  | 22.6  | 25.1  | 27.6  | 31.7  | 34.1  | 35.1  | 33.3  | 28.7  | 23.4  | 18.6 |
|                                                                                  |             | MCWB   | 9.8   | 10.4  | 12.5  | 13.3  | 15.9  | 18.0  | 19.6  | 19.3  | 18.4  | 14.9  | 11.3  | 9.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.3  | 15.5  | 17.4  | 17.8  | 19.7  | 21.6  | 22.9  | 23.6  | 22.8  | 20.5  | 16.2  | 15.4 |
|                                                                                  |             | MCDB   | 19.1  | 19.7  | 25.6  | 27.9  | 32.7  | 33.9  | 34.4  | 33.3  | 33.8  | 26.5  | 20.4  | 17.6 |
|                                                                                  | 2%          | WB     | 13.8  | 14.0  | 15.8  | 16.4  | 18.5  | 20.2  | 21.9  | 22.2  | 21.7  | 18.8  | 15.0  | 13.7 |
|                                                                                  |             | MCDB   | 17.9  | 19.2  | 23.1  | 25.9  | 29.5  | 32.2  | 33.8  | 34.4  | 32.5  | 26.8  | 22.1  | 17.1 |
|                                                                                  | 5%          | WB     | 12.7  | 12.9  | 14.8  | 15.2  | 17.5  | 19.4  | 21.2  | 21.3  | 20.7  | 17.6  | 14.0  | 12.4 |
|                                                                                  |             | MCDB   | 17.0  | 18.6  | 21.6  | 23.7  | 27.6  | 30.7  | 32.8  | 33.5  | 30.4  | 25.1  | 21.3  | 15.9 |
|                                                                                  | 10%         | WB     | 11.7  | 12.0  | 13.7  | 14.2  | 16.6  | 18.6  | 20.5  | 20.4  | 19.8  | 16.5  | 13.2  | 11.2 |
|                                                                                  |             | MCDB   | 16.4  | 16.9  | 19.9  | 21.7  | 25.3  | 29.0  | 31.6  | 32.2  | 29.6  | 23.6  | 20.0  | 15.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.4  | 13.6  | 15.0  | 15.6  | 15.1  | 16.8  | 17.2  | 17.6  | 16.8  | 16.4  | 16.1  | 14.3 |
|                                                                                  |             | MCDBR  | 19.7  | 20.5  | 20.6  | 21.6  | 20.9  | 20.6  | 18.9  | 19.2  | 19.7  | 21.8  | 21.5  | 19.5 |
|                                                                                  |             | MCWBR  | 9.4   | 9.5   | 9.3   | 8.7   | 7.8   | 7.2   | 6.5   | 6.8   | 6.8   | 8.3   | 9.5   | 9.4  |
|                                                                                  | 5% WB       | MCDBR  | 13.6  | 14.4  | 16.9  | 17.5  | 17.9  | 18.2  | 16.9  | 17.1  | 16.3  | 16.1  | 16.4  | 12.6 |
|                                                                                  |             | MCWBR  | 7.6   | 7.9   | 8.4   | 7.8   | 6.9   | 6.5   | 6.0   | 6.3   | 5.9   | 6.6   | 8.0   | 7.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.290  | 0.299 | 0.319 | 0.333 | 0.346 | 0.347 | 0.375 | 0.359 | 0.341 | 0.327 | 0.297 | 0.298 |      |
|                                                                                  | taud        | 2.573  | 2.558 | 2.482 | 2.449 | 2.444 | 2.454 | 2.437 | 2.467 | 2.503 | 2.515 | 2.588 | 2.578 |      |
|                                                                                  | Ebn at Noon | 934    | 958   | 958   | 953   | 938   | 932   | 904   | 916   | 923   | 912   | 918   | 902   |      |
|                                                                                  | Edh at Noon | 78     | 88    | 103   | 112   | 114   | 113   | 114   | 109   | 100   | 91    | 77    | 74    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.25   | 4.16  | 5.59  | 6.80  | 7.47  | 8.02  | 7.65  | 7.22  | 6.18  | 4.73  | 3.62  | 2.95  |      |
|                                                                                  | RadStd      | 0.22   | 0.40  | 0.27  | 0.29  | 0.38  | 0.39  | 0.36  | 0.20  | 0.20  | 0.28  | 0.17  | 0.20  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | -1.05   | -1.98  | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (52 neighbors) |  | +0.63 | N/A     | -1.69  | +0.70 | N/A     | N/A   | N/A | -109        | +222    | +114    |         |

Nomenclature: See separate page